

Pricing a house in Aarhus

Machine Learning in Finance · Session 7

The house

150 m²

living area

1973

year built

5

rooms

villa

type

Aarhus V

postal town

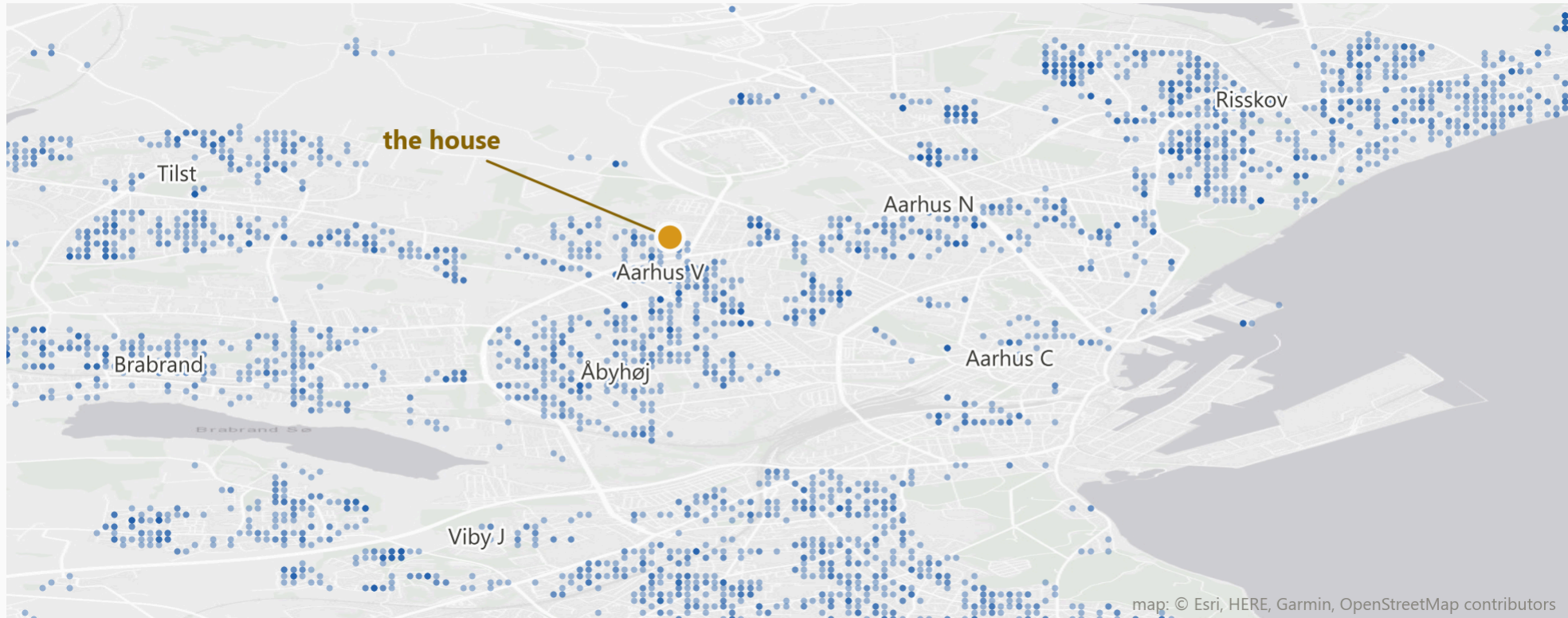
3.6 km

from the
cathedral

A buyer has found this house and asks a bank for a mortgage. Before the bank lends, it needs one number: what the house is worth. It sold in January 2024.

Where it stands

Every house sold in the inner Aarhus postcodes, 2021 to 2024



Each blue dot is a sale in the table, placed to within about a hundred metres. The [live map](#) zooms to any street in the table.

The street

Drag to look around, and use the arrows to walk down the street. If the frame stays blank, open it in [Street View](#).

Write down a price

150 m²

living area

1973

year built

5

rooms

villa

type

Aarhus V

postal town

3.6 km

from the
cathedral

You now have what a buyer has before the viewing: the size, the age, the rooms, the town, the distance to the centre, and the street. What did it sell for in January 2024?

Your price kroner

The number stays in this browser and comes back in Part 6, next to the model's valuation and the sale price.

What the bank may lend

§ 5 · the limit

“Inden for en lånegrænse på 80 pct. af ejendommens værdi kan der ydes lån til følgende ejendoms kategorier: 1) Ejerboliger til helårsbrug.”

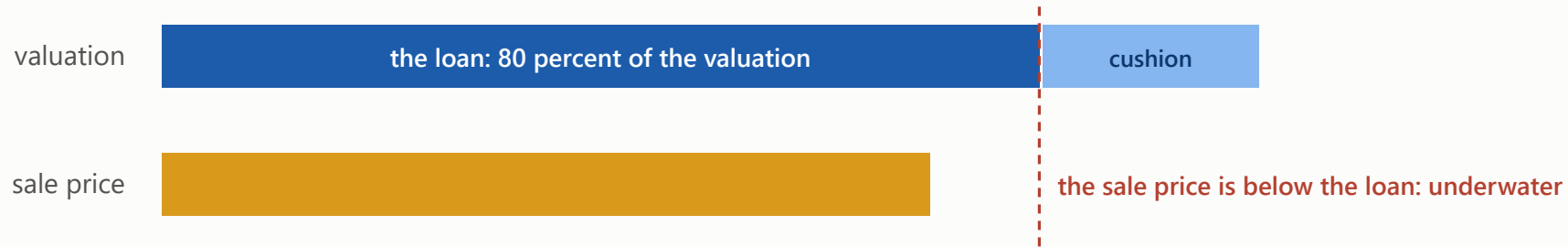
§ 10 · the value

“Realkreditinstituttet skal ansætte en kontantværdi af ejendommen til brug for låneudmålingen. Værdien skal ligge inden for det skønnede beløb, som ejendommen kan handles til ... (markedsværdi).”

The bank may lend 80 percent of the value, and must set that value itself, as the price the house would fetch in an ordinary sale.

Lov om realkreditlån og realkreditobligationer m.v., lovbekendtgørelse nr. 1541 (2025), §§ 5 and 10.

What the lender risks



a valuation more than a fifth above the sale price puts the loan underwater on the day it is written

Two numbers score every rule today: the error in kroner, and the share of sales on which a loan of 80 percent of the rule's value would have exceeded the price.

Who values with a model

The mortgage banks

may replace the visit to an owner-occupied home with “en dokumenteret værdiansættelsesmodel”, and may monitor the value of every home they have lent against with one.

A coefficient, in its words

“Fra data om salg af ejendomme i hele landet ved vi, hvor meget forskellige forhold gennemsnitligt påvirker salgsprisen på en ejendom.”

Bekendtgørelse nr. 807 (2022) om værdiansættelse af pant og lån i fast ejendom, §§ 5 and 33 · vurdst.dk and vurderingsportalen.dk · Zillow Group, Form 8-K, 2 November 2021.

The tax authority

“bruger et system, der bygger på en matematisk model, til at vurdere landets 2,4 mio. ejendomme”. Its inputs are the living area, the year built, the surroundings and “salgspriser for ejendomme i området”.

When it goes wrong

Zillow wound down its house-buying business in 2021 after its model overpaid: a 304 million dollar write-down in one quarter. “The unpredictability in forecasting home prices far exceeds what we anticipated.”

Today, in six steps

Part 1	Look at the market: the file, one column, a mask, a number per town, a map and a scatter	pandas · matplotlib
Part 2	Rules by hand: a price per square metre as a loop, as one line, as a function, and over time	loops · dicts · def · a line
Part 3	A first model: the sealed 2024 sales, two rules that need no model, a fitted line	fit · predict · RMSE
Part 4	Choose columns without touching the 2024 sales	KFold · cross_val_score
Part 5	Location as text turned into numbers, a penalty chosen by grid search, the log	get_dummies · GridSearchCV
Part 6	Value the house, renovate it, move it across town, and find where the model fails	predict · copy · a loop

Nothing here is new. Open the deck on your own laptop: every cell runs and can be edited. A slide marked **PLAY** holds a working cell to change as you like; a slide marked **YOUR TURN** has its answer on the slide after.

The data

7,621

house sales

22

postal towns

15 quarters

Jan 2021 to Sep 2024

13

columns

These are ordinary sales of villas and townhouses in the Aarhus postcodes, with family transfers and auctions left out, from the public sale records as collected by boliga.dk. Each house is placed by the national address register; addresses are dropped and locations rounded to about a hundred metres.

1 • Looking at the market

PART 1

The file, one column, a question asked of every row, a number per town, a map and a scatter.

Reading the file

Python Code

```
houses = pd.read_csv("data/aarhus_houses.csv")
```

```
print(houses.shape)
```

```
houses.head()
```

(7621, 13)

	price	area	rooms	built	type	town	zip	centre_km	sold	quarter	date	lat	lon
0	7995000	222	5	2012	villa	Højbjerg	8270	5.7	2021	2021Q1	2021-01-01	56.107	10.186
1	4190000	191	5	2009	villa	Lystrup	8520	9.2	2021	2021Q1	2021-01-01	56.239	10.193
2	3000000	171	5	1980	villa	Egå	8250	6.6	2021	2021Q1	2021-01-01	56.207	10.267
3	1500000	160	5	2021	villa	Malling	8340	14.8	2021	2021Q1	2021-01-03	56.024	10.185
4	2200000	84	4	1981	townhouse	Aarhus V	8210	3.6	2021	2021Q1	2021-01-03	56.171	10.158

The columns

price	the sale price, in kroner	zip	the postcode
area	living area, m ²	centre_km	kilometres from the cathedral
rooms	number of rooms	sold · quarter · date	when the sale happened
built	the year the house was built	lat · lon	the location, to about 100 metres
type · town	villa or townhouse; one of 22 postal towns		

Six of the columns describe the house. **price** is what the model has to produce, **sold** decides which rows it may learn from, and **lat** and **lon** are for the map.

One column

Python Code

```
prices = houses['price']  
  
print(round(prices.mean()))  
print(prices.median())  
print(prices.max())
```

```
4098945  
3645000.0  
43300000
```

A column on its own is a Series with its own methods. The mean sale is 450,000 kroner above the median: a few very expensive houses pull it up, and the most expensive of all sold for 43.3 million.

The most expensive sales

PLAY

Python Code

```
houses.sort_values('price', ascending=False).head(3)
```

	price	area	rooms	built	type	town	zip	centre_km	sold	quarter	date	lat	lon
3574	43300000	594	8	2006	villa	Risskov	8240	5.8	2022	2022Q2	2022-06-15	56.198	10.268
2579	30100000	433	12	1907	villa	Aarhus C	8000	2.6	2022	2022Q1	2022-01-13	56.133	10.209
6248	19200000	325	10	1915	villa	Højbjerg	8270	5.7	2024	2024Q1	2024-01-11	56.106	10.227

Sort by a column, then take the top of the table. Change the cell: the cheapest three, the oldest, the largest, the newest, or the most expensive in your own town.

A question asked of every row

Python Code

```
chosen = houses[houses['town'] == 'Risskov']  
  
print(len(chosen))  
print(chosen['price'].median())
```

```
820  
5050000.0
```

The comparison inside the brackets is asked of every row, and only the rows that answer **True** are kept: 820 sales in Risskov, with a median of 5.05 million kroner.

A number per town

Python Code

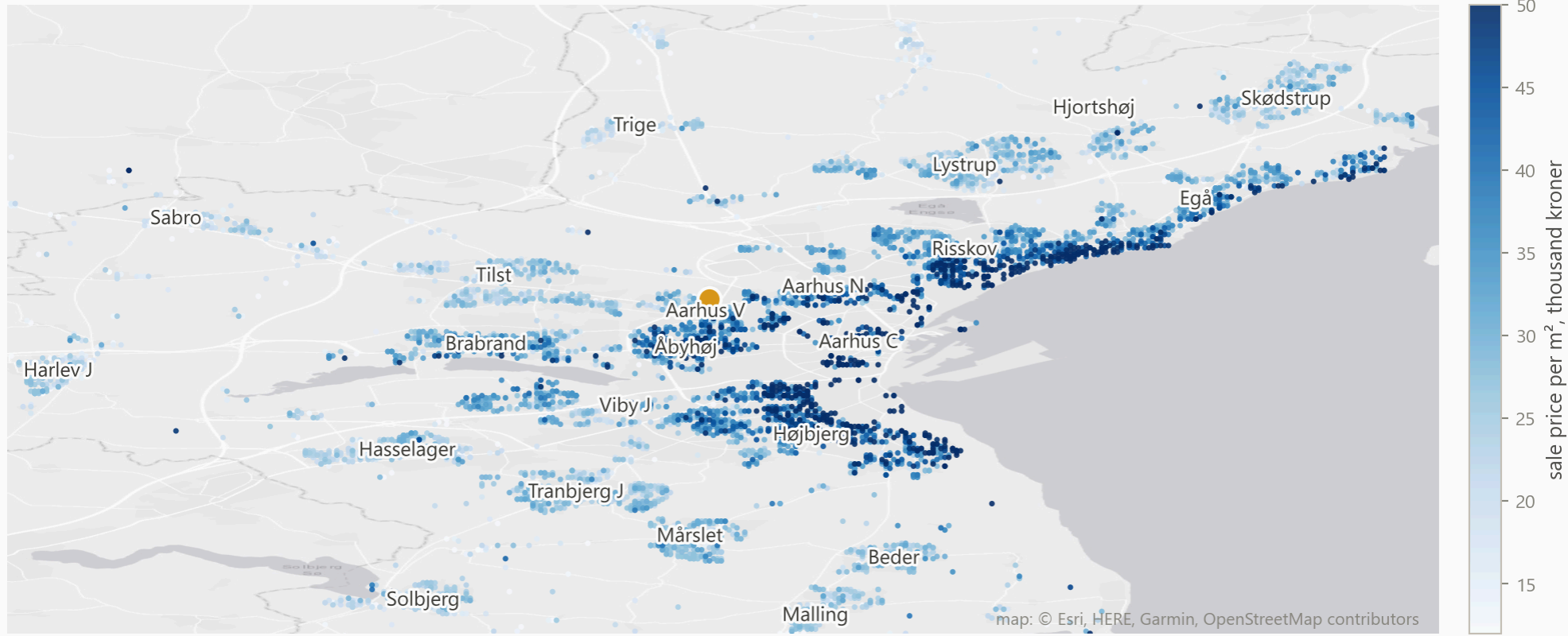
```
by_town = houses.groupby('town')['price'].median().sort_values()

print(by_town.head(3))
print(by_town.tail(3))
```

```
town
Trige      2450000.0
Sabro      2662500.0
Tranbjerg J 2750000.0
Name: price, dtype: float64
town
Risskov    5050000.0
Højbjerg   5325000.0
Aarhus C   6300000.0
```

Where Aarhus is expensive

7,621 sales, coloured by the price per square metre; the house in amber

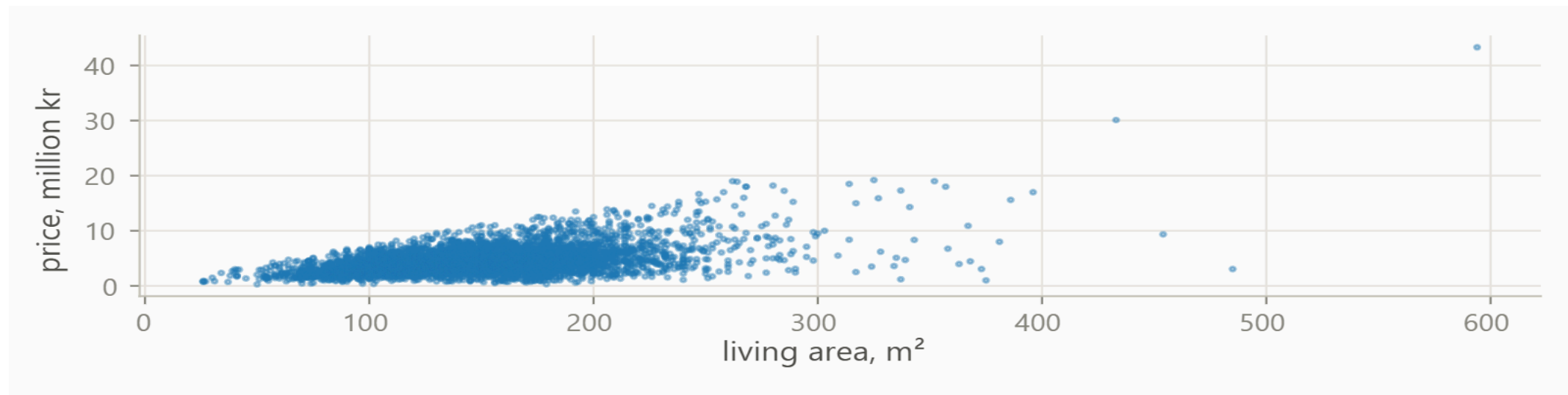


The price per square metre runs from 18,900 kroner in Sabro to 52,400 in Aarhus C, and it falls with the distance from the centre and rises towards the bay.

Drawing the market PLAY

Python Code

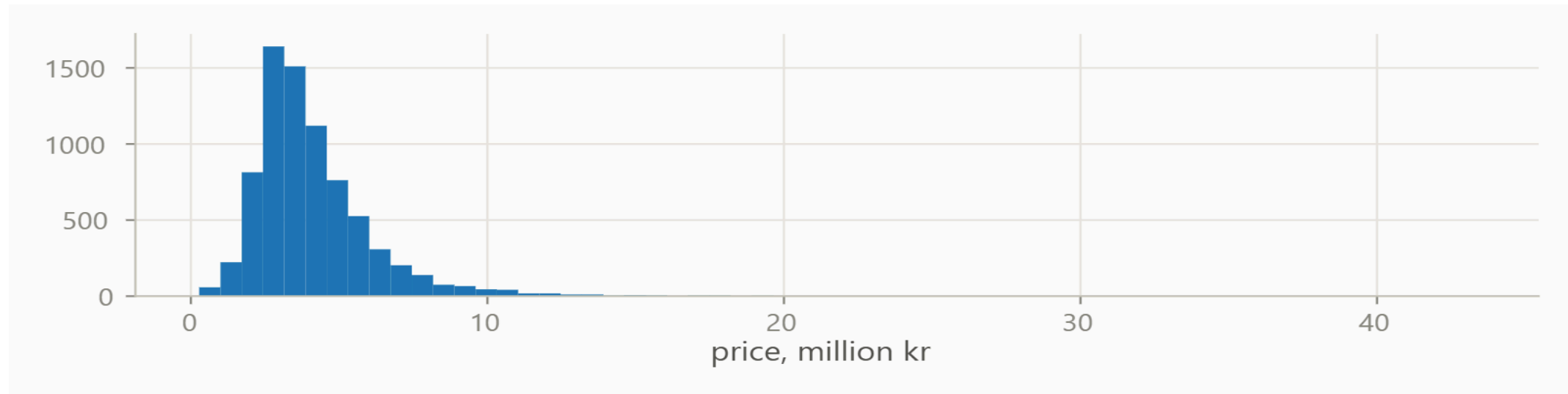
```
fig, ax = plt.subplots(figsize=(8, 2.4))
ax.scatter(houses['area'], houses['price'] / 1000000, s=4, alpha=0.4)
ax.set_xlabel("living area, m2")
ax.set_ylabel("price, million kr")
fig.tight_layout()
plt.show()
```



The shape of the prices

Python Code

```
fig, ax = plt.subplots(figsize=(8, 2.4))  
ax.hist(houses['price'] / 1000000, bins=60)  
ax.set_xlabel("price, million kr")  
fig.tight_layout()  
plt.show()
```



2 • Rules by hand

PART 2

The price per square metre, the number every Danish listing quotes: as a loop, as one line, as a function, and quarter by quarter.

One house, by hand

Python Code

```
first = houses.iloc[0]  
  
print(first['price'] / first['area'])
```

```
36013.51351351351
```

`.iloc[0]` picks the first row: a 222 square metre villa in Højbjerg that sold for 7,995,000 kroner, which is 36,014 kroner per square metre.

Every house, with a loop

Python Code

```
per_m2 = []
for i in range(len(houses)):
    row = houses.iloc[i]
    per_m2.append(row['price'] / row['area'])

print(len(per_m2))
print(per_m2[:3])
```

```
7621
[np.float64(36013.51351351351), np.float64(21937.17277486911),
np.float64(17543.859649122805)]
```

This is the accumulator pattern: an empty list, then one row per turn of the loop and one **append**. The loop runs 7,621 times.

Every house, in one line

Python Code

```
houses['per_m2'] = houses['price'] / houses['area']
```

```
print(houses['per_m2'].head(3))
```

```
print(houses['per_m2'].median())
```

```
0      36013.513514
```

```
1      21937.172775
```

```
2      17543.859649
```

```
Name: per_m2, dtype: float64
```

```
27321.428571428572
```

Dividing one column by another divides row by row, and the result is a new column. The loop and the one line give the same numbers; the one line runs the loop in

A dictionary, one entry per year

Python Code

```
sales = {}  
for year in [2021, 2022, 2023, 2024]:  
    sales[year] = len(houses[houses['sold'] == year])  
  
print(sales)
```

```
{2021: 2532, 2022: 1834, 2023: 1846, 2024: 1409}
```

A mask runs inside the loop, and a dictionary collects the answers by name. The same shape of loop can collect anything: a count, a median, a share.

Your turn 1

In 2022 the yield on Danish mortgage bonds went from 1.5 to 3.7 percent, and on to 4.8 in 2023. Fill a dictionary with the median price per square metre in each year, then get the same numbers in one line with **groupby**.

Python Code

```
medians = {}
for year in [2021, 2022, 2023, 2024]:
    medians[year] = ...
print(medians)

print(houses.groupby(...)[...].median())
```

Your turn 1: answer

```
medians = {}  
for year in [2021, 2022, 2023, 2024]:  
    medians[year] = houses[houses['sold'] == year]['per_m2'].median()  
print(medians)  
# {2021: 27034.0, 2022: 27544.0, 2023: 27577.0, 2024: 27083.0}  
  
print(houses.groupby('sold')['per_m2'].median())
```

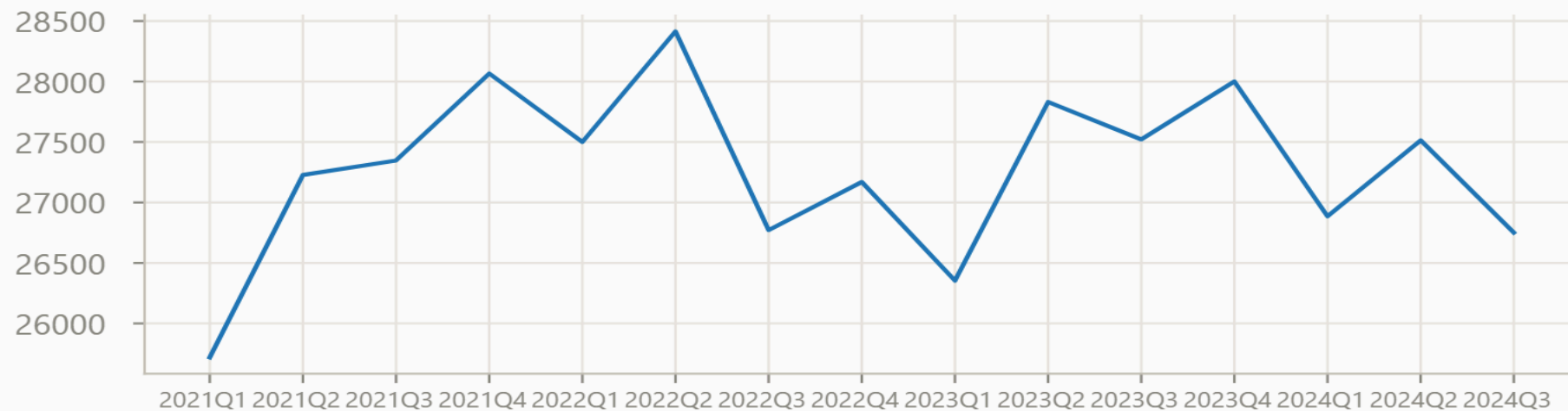
Mortgage rates tripled, and the median price per square metre in Aarhus moved by 500 kroner. Fewer houses changed hands: 2,532 sales in 2021 and 1,834 in 2022.

The market over time

PLAY

Python Code

```
by_quarter = houses.groupby('quarter')['per_m2'].median()  
fig, ax = plt.subplots(figsize=(8, 2.4))  
ax.plot(by_quarter.index, by_quarter.values)  
ax.tick_params(axis='x', labelsiz=8)  
plt.show()
```



A function that describes a house

Python Code

```
def describe(row):  
    return f"{row['area']} m² {row['type']}, built {row['built']}, {row['town']}: {row['price']: ,.0f} kr"  
  
print(describe(houses.iloc[0]))
```

```
222 m² villa, built 2012, Højbjerg: 7,995,000 kr
```

The function takes one parameter, builds one f-string and returns it. The comma in `: ,.0f` groups the thousands.

Your turn 2

Describe the three most expensive sales, one line each, using `describe` inside a loop.

Python Code

```
top = houses.sort_values('price', ascending=False)
for i in range(3):
    ...
```

Your turn 2: answer

```
top = houses.sort_values('price', ascending=False)
for i in range(3):
    print(describe(top.iloc[i]))
```

```
594 m² villa, built 2006, Risskov: 43,300,000 kr
433 m² villa, built 1907, Aarhus C: 30,100,000 kr
325 m² villa, built 1915, Højbjerg: 19,200,000 kr
```

A function written once is called three times by the loop. Calling it for all 7,621 houses would change only the number in `range`.

3 • A first model

PART 3

The rows to learn from, two rules that need no model, and a fitted line: a split, an error in kroner, `fit` and `predict`.

Two functions for scoring a rule

Python Code

```
def rmse(actual, predicted):  
    return np.sqrt(((actual - predicted) ** 2).mean())  
  
def underwater(actual, predicted):  
    return (0.8 * predicted > actual).mean()
```

The first is the root mean squared error, in kroner. The second compares 80 percent of every valuation with the price the house sold for; the comparison is a mask, and the mean of a mask is a share.

The rows to learn from, and the rows to check on

Python Code

```
train = houses[houses['sold'] <= 2023]
test = houses[houses['sold'] == 2024]

print(len(train), len(test))
```

```
6212 1409
```

A bank valuing a house in 2024 has the sales up to 2023. The 1,409 sales of 2024 stay sealed until a rule has been chosen, and then score it once.

Rule one: every house is worth the average

Python Code

```
guess = train['price'].mean()

print(round(guess))
print(rmse(test['price'], guess))
print(underwater(test['price'], guess))
```

```
4109721
1905702.1237462144
0.4038325053229241
```

The rule is off by 1.9 million kroner on a typical house and underwater on 40 percent of the 2024 sales. Every other rule has to beat these two numbers.

Rule two: a price per square metre

Python Code

```
rate = (train['price'] / train['area']).mean()

print(rate)
print(rmse(test['price'], rate * test['area']))
print(underwater(test['price'], rate * test['area']))
```

```
29293.939197278203
1634510.280159343
0.29595457771469125
```

This is the rule of thumb from Part 2, computed on the training rows only. It cuts the error to 1.63 million and the underwater share to 30 percent.

Fitting a line

Python Code

```
from sklearn.linear_model import LinearRegression

model = LinearRegression()
model.fit(train[['area']], train['price'])

print(model.intercept_, model.coef_)
```

```
234055.96749066375 [27325.87987818]
```

The three steps are import, create and fit. The line starts at 234,000 kroner and adds 27,326 for every square metre.

Predicting

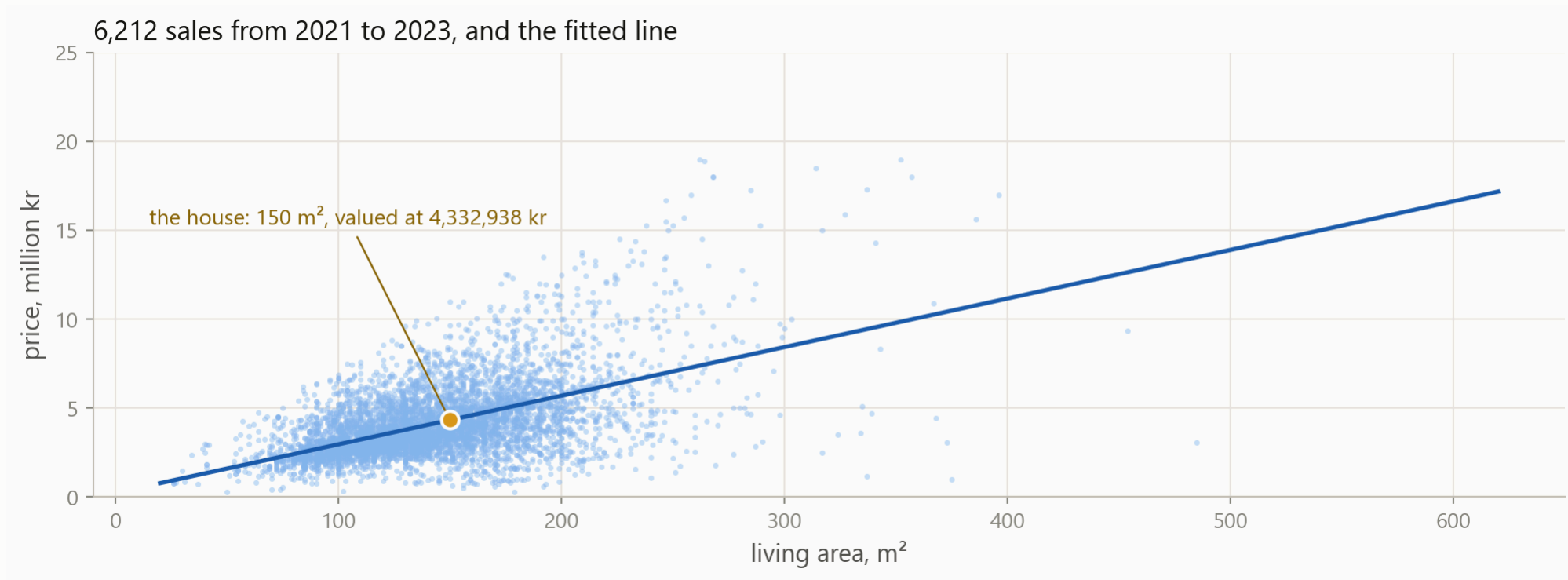
Python Code

```
predicted = model.predict(test[['area']])  
  
print(rmse(test['price'], predicted))  
print(underwater(test['price'], predicted))
```

```
1629762.7842889985  
0.2945351312987935
```

The error is 1,629,763 kroner, 5,000 better than the rule of thumb. Area on its own explains part of the spread in prices, and a straight line through it cannot do more.

The line, drawn



Every dot is a sale in the training rows. The line values the house from the first slide at 4,332,938 kroner; the dots at 150 square metres run from 1.5 to 8 million.

One column at a time PLAY

Python Code

```
other = LinearRegression()
other.fit(train[['centre_km']], train['price'])

print(other.coef_)
print(rmse(test['price'], other.predict(test[['centre_km']])))
print(underwater(test['price'], other.predict(test[['centre_km']])))
```

```
[-215519.9156758]
1746646.3250655967
0.36621717530163234
```

On its own, every kilometre from the cathedral takes 215,520 kroner off. Swap the column for **built** or **rooms**, then put two columns in the list at once. Which pair gets the error below 1.4 million?

Area, and the distance to the centre

Python Code

```
pair = LinearRegression()
pair.fit(train[['area', 'centre_km']], train['price'])

print(pair.coef_)
print(rmse(test['price'], pair.predict(test[['area', 'centre_km']])))
print(underwater(test['price'], pair.predict(test[['area', 'centre_km']])))
```

```
[ 29611.42412827 -249003.19239279]
1389709.9460118285
0.24627395315826828
```

Together the two columns bring the error to 1.39 million and the underwater share to 25 percent. Every kilometre from the cathedral now takes 249,000 kroner off with the

4 • Choosing columns

PART 4

Which of the numeric columns belong in the rule, decided without touching the 2024 sales: `KFold`, `cross_val_score`, and a loop over candidates.

The folds

Python Code

```
from sklearn.model_selection import KFold, cross_val_score  
  
folds = KFold(n_splits=5, shuffle=True, random_state=0)
```

Trying each set of columns on the 2024 sales and keeping the best would use the test rows up. Instead the training rows are cut into five blocks, each block takes a turn as the validation block, and the five errors are averaged. Houses are interchangeable rows, so a shuffled **KFold** is right here; a time series needed **TimeSeriesSplit**.

Candidates, in a loop PLAY

Python Code

```
candidates = {
    'area': ['area'],
    'area, built': ['area', 'built'],
    'area, built, centre_km': ['area', 'built', 'centre_km'],
}

cv_error = {}
for name in candidates:
    scores = cross_val_score(LinearRegression(), train[candidates[name]], train['price'],
                             cv=folds, scoring='neg_root_mean_squared_error')
    cv_error[name] = round(-scores.mean())
print(cv_error)
```

```
{'area': 1781237, 'area, built': 1775503, 'area, built, centre_km': 1509031}
```

Candidates go in and errors come out: **built** takes 6,000 kroner off and the distance to the centre 270,000. Add your own. Does **rooms** help?

The chosen rule, on the 2024 sales

Python Code

```
three = ['area', 'built', 'centre_km']

chosen = LinearRegression()
chosen.fit(train[three], train['price'])

print(rmse(test['price'], chosen.predict(test[three])))
print(underwater(test['price'], chosen.predict(test[three])))
```

```
1386595.399125231
0.23988644428672817
```

The rule scores 1.39 million kroner and 24 percent underwater. The 2024 sales were opened once, after the choosing was done, so these two numbers are a fair report.

Three coefficients

Python Code

```
for name, b in zip(three, chosen.coef_):  
    print(name, round(b))
```

```
area 29624  
built 2138  
centre_km -253683
```

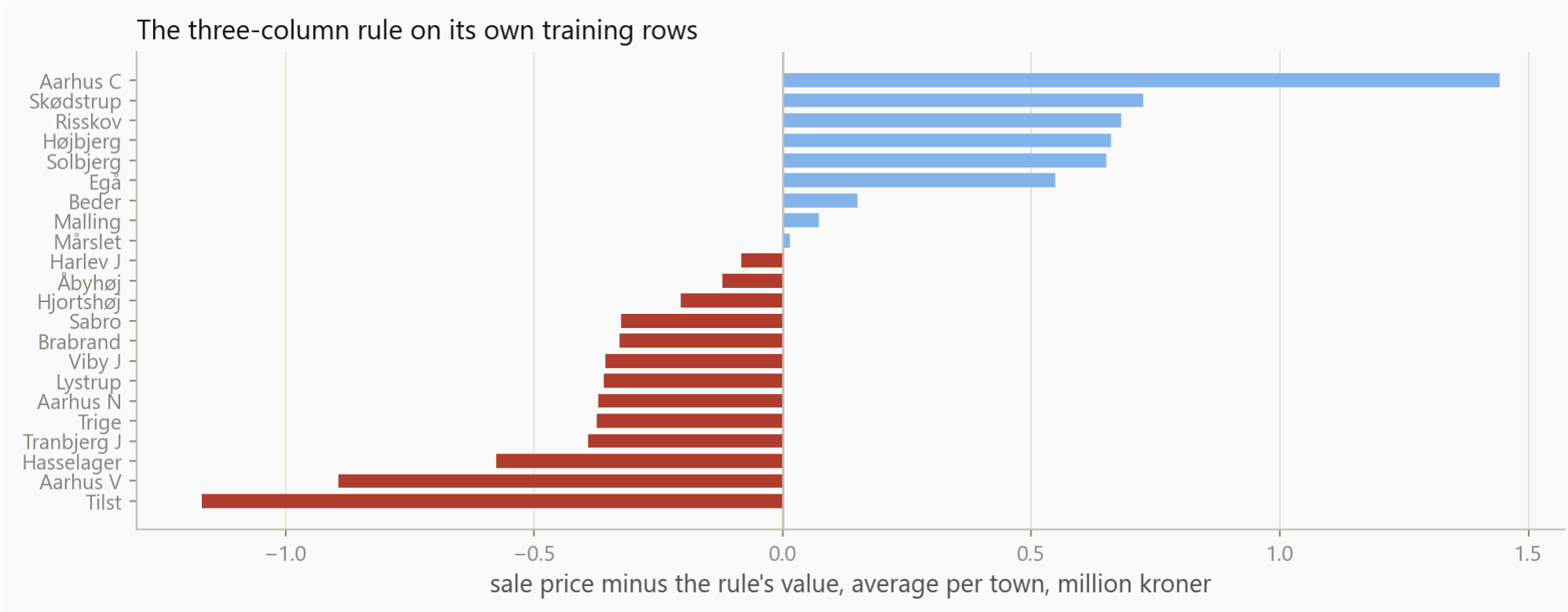
A square metre is worth 29,624 kroner and a year of age costs 2,138, each with the other columns held fixed. Every kilometre from the cathedral takes 253,683 kroner off.

5 • Location, and a penalty

PART 5

What the three columns cannot see, two columns of text turned into numbers, a penalty chosen by grid search, and the log.

What the three columns cannot see



On its own training rows the rule undervalues Aarhus C by 1.4 million kroner on average and overvalues Tilst by 1.1 million. The distance to the cathedral is not the whole of location.

Text into numbers

Python Code

```
wide = pd.get_dummies(houses, columns=['town', 'type'], dtype=int)

wtrain = wide[wide['sold'] <= 2023]
wtest = wide[wide['sold'] == 2024]

columns = []
for c in wide.columns:
    if c not in ['price', 'sold', 'quarter', 'date', 'zip', 'lat', 'lon', 'per_m2']:
        columns.append(c)
print(len(columns))
```

28

Two columns of text became twenty-four columns of zeros and ones, one per town and one per type. The loop collects every column except the target and the ones that only identify the sale.

All twenty-eight, unpenalised

Python Code

```
everything = LinearRegression()  
everything.fit(wtrain[columns], wtrain['price'])  
  
print(rmse(wtrain['price'], everything.predict(wtrain[columns])))  
print(rmse(wtest['price'], everything.predict(wtest[columns])))  
print(underwater(wtest['price'], everything.predict(wtest[columns])))
```

```
1378814.2543575515  
1247050.684145275  
0.1880766501064585
```

The error on the 2024 sales is 1.25 million kroner, 140,000 better than the three columns, with 19 percent underwater. Training and test errors are close: 6,212 rows against 28 columns leave little room to overfit.

A penalty, chosen by grid search

Python Code

```
from sklearn.linear_model import Ridge, Lasso
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import Pipeline
from sklearn.model_selection import GridSearchCV
pipe = Pipeline([('scale', StandardScaler()), ('ridge', Ridge())])
grid = {'ridge__alpha': [0.01, 0.1, 1, 10, 100, 1000]}
search = GridSearchCV(pipe, grid, cv=folds, scoring='neg_root_mean_squared_error')
search.fit(wtrain[columns], wtrain['price'])
print(search.best_params_, -search.best_score_)
print(rmse(wtest['price'], search.predict(wtest[columns])))
```

```
{'ridge__alpha': 10} 1389834.4909856953
1246637.3931116923
```

The grid picks alpha 10, and the test error moves by 400 kroner. With two hundred rows per column the penalty has little to do; with nineteen near-copies of one volatility it had a lot.

What the lasso keeps

PLAY

Python Code

```
lasso = Pipeline([('scale', StandardScaler()), ('lasso', Lasso(alpha=100000, max_iter=50000))])
lasso.fit(wtrain[columns], wtrain['price'])

kept = []
for name, b in zip(columns, lasso.named_steps['lasso'].coef_):
    if b != 0:
        kept.append(name)
print(len(kept))
print(rmse(wtest['price'], lasso.predict(wtest[columns])))
```

```
12
1293409.7059645178
```

At alpha 100,000 the lasso keeps twelve of the twenty-eight columns: **area**, **centre_km**, the type and eight of the towns, with **built** set to exactly zero. Raise alpha until five remain, then print **kept** to see which.

The log of the price

Python Code

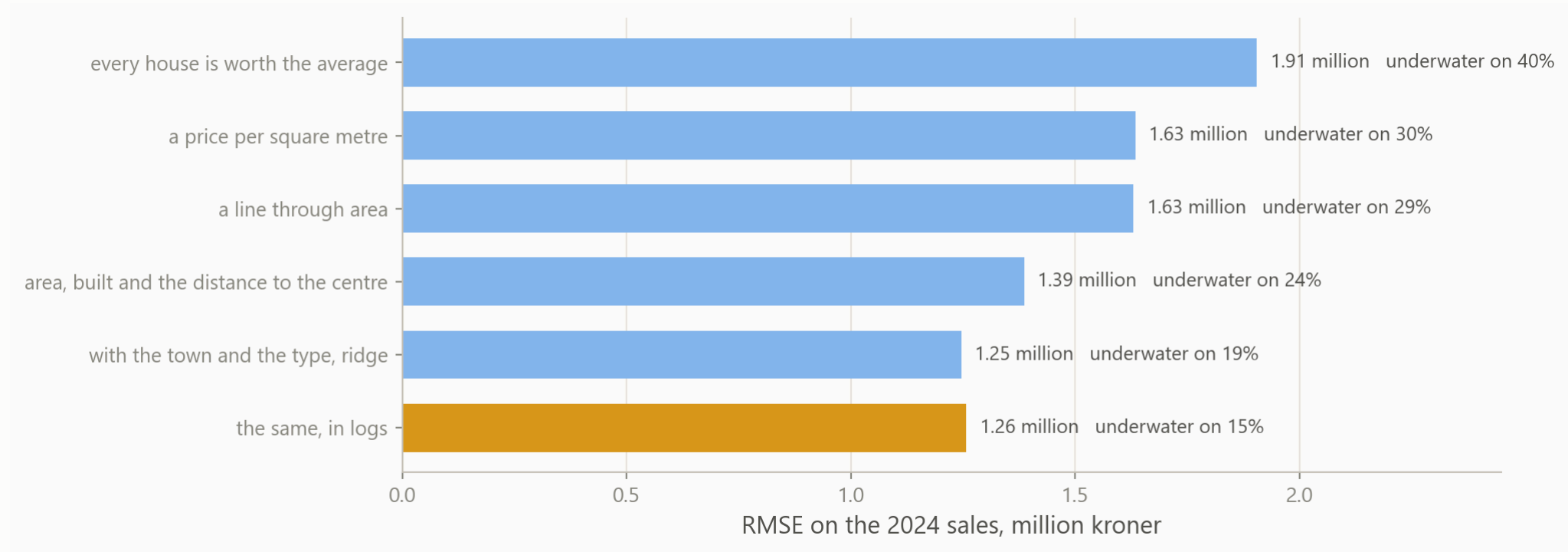
```
valuation = GridSearchCV(pipe, grid, cv=folds, scoring='neg_root_mean_squared_error')
valuation.fit(wtrain[columns], np.log(wtrain['price']))

valued = np.exp(valuation.predict(wtest[columns]))
print(rmse(wtest['price'], valued))
print(underwater(wtest['price'], valued))
```

```
1257437.3778117676
0.15117104329311568
```

This is the same grid search on `np.log(price)`, with the predictions brought back by `np.exp`. In logs the model fits percentages rather than kroner: the error in kroner is 11,000 worse and the underwater share falls from 19 to 15 percent. This is the rule a lender would keep.

The error, rule by rule



Six rules were scored on the same 2024 sales. The share of loans underwater fell from 40 percent to 15, and no tool used on the way was new today.

What is missing

The house itself

bathrooms, the plot, the basement, the roof, the heating, the year of the last renovation. All of it is in the building register, BBR, which the tax authority's model reads and this table does not.

The condition

two houses of the same size and age on one street sell far apart when one has a new kitchen and the other a leaking roof. No register holds that.

Six columns are off by 15 percent on a typical house. The tax authority's model has dozens of columns, and a bank still sends someone to look.

6 • One house

PART 6

The house from the first slide, valued against your guess and its sale price, renovated, moved across town, and the sales the model gets most wrong.

A valuation function

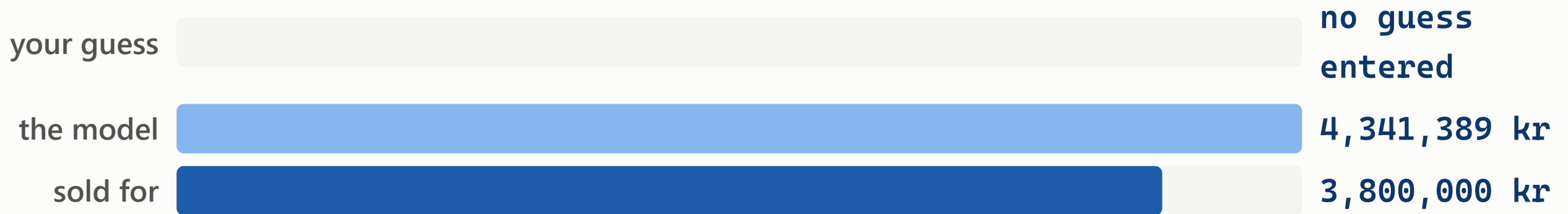
Python Code

```
def value(row):  
    return np.exp(valuation.predict(row[columns]))[0]  
  
house = wtest[wtest['town_Aarhus V'] == 1].head(1)  
print(value(house))
```

4341389.378464201

The mask keeps the 2024 sales in Aarhus V and `.head(1)` the first of them, as a table with one row, which is what `.predict()` needs. That is the house from the first slide: the model values it at 4,341,389 kroner, and a bank lending 80 percent would offer 3,473,112.

The price



No guess was entered on the Write down a price slide.

The model was 541,389 kroner above the price, 14 percent.

The house sold on 20 January 2024 for 3,800,000 kroner.

A typical miss

Python Code

```
print(describe(houses.loc[6283]))
```

```
150 m² villa, built 1973, Aarhus V: 3,800,000 kr
```

The model is 14 percent high, and that is a typical miss: the rule is within 10 percent of the price on a third of the 2024 sales and within 20 percent on two thirds. The loan, 3,473,112, was still below the price.

Renovate it

PLAY

Python Code

```
bigger = house.copy()
bigger['area'] = 180

print(value(bigger) - value(house))
```

639336.2144045141

Thirty square metres more are worth 639,336 kroner to the model; `.copy()` first, so the original row stays as it is. Try a sixth room, a build year of 2000, two kilometres closer to the cathedral, or a townhouse (`type_villa` 0, `type_townhouse` 1).

The same house, in every town

Python Code

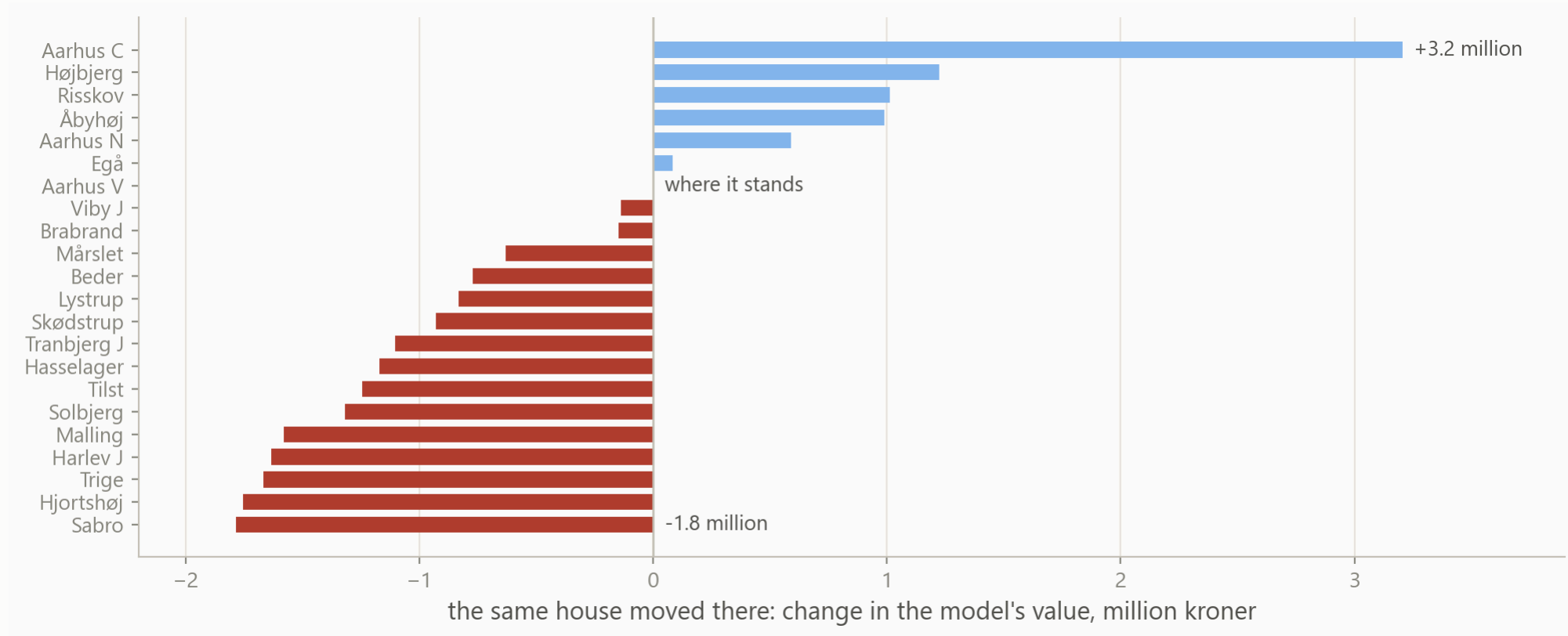
```
typical_km = train.groupby('town')['centre_km'].median()

moved_value = {}
for name in houses['town'].unique():
    moved = house.copy()
    for c in columns:
        if 'town_' in c:
            moved[c] = 0
    moved['town_' + name] = 1
    moved['centre_km'] = typical_km[name]
    moved_value[name] = round(value(moved))
print(min(moved_value.values()), max(moved_value.values()))
```

2619327 7607051

The inner loop clears every town column, the next line switches one on, and the house moves to that town's typical distance. It is worth 2.6 million kroner in Sabro and 7.6 million in Aarhus C.

The location premium, drawn



Size, age and type are held fixed, so the bars differ only by the town and its distance from the centre. Between Sabro and Aarhus C the gap is 5.0 million kroner.

Where the model is most wrong PLAY

Python Code

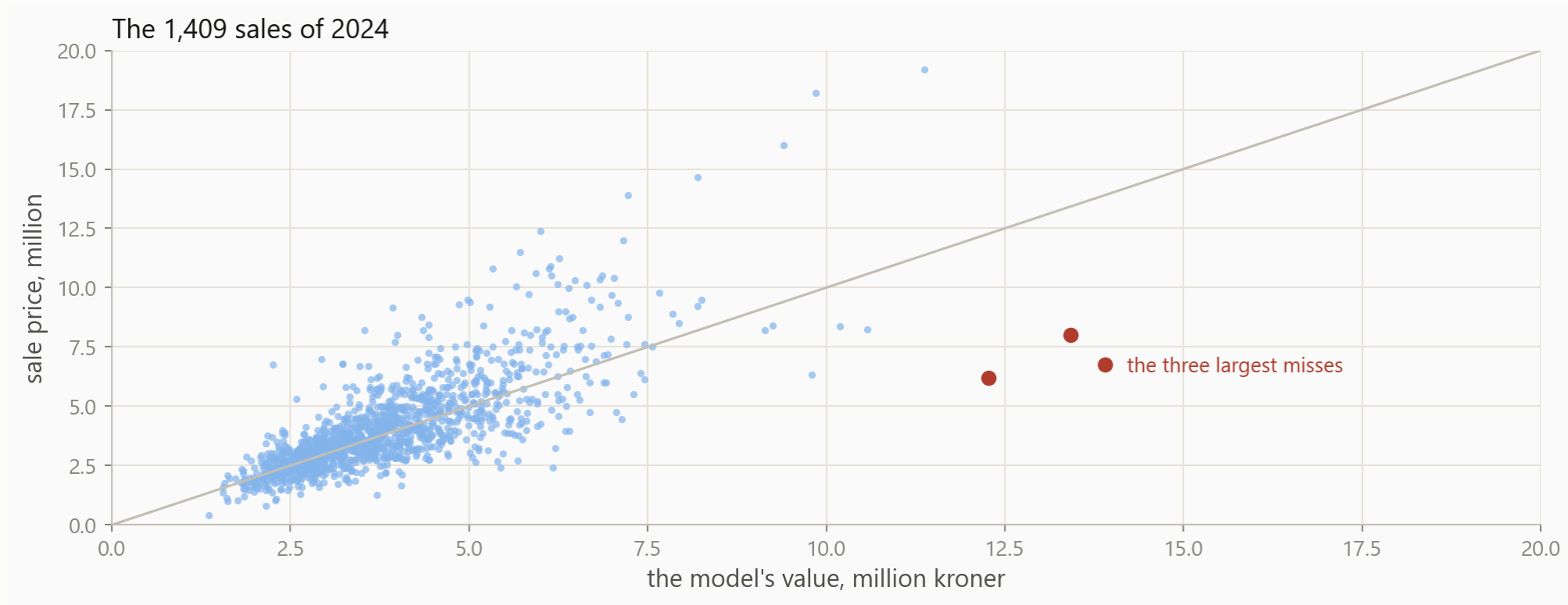
```
errors = wtest['price'] - np.exp(valuation.predict(wtest[cOLUMNS]))
worst = errors.sort_values().head(3)

houses.loc[worst.index][['price', 'area', 'rooms', 'built', 'town']]
```

	price	area	rooms	built	town
6730	6750000	358	12	1898	Aarhus N
7258	6200000	328	13	1900	Aarhus N
6299	8000000	381	12	2023	Egå

The three largest misses are houses of 330 to 380 square metres, valued at 12 to 14 million and sold for 6 to 8. Look at the other end with `.tail(3)`: which houses sold for far more than the model's value?

Valued against sold



A dot on the grey line is a house valued at exactly its price. The cloud widens as prices rise: above 8 million the six columns say less and less about what a buyer paid.

A letter from the bank

Python Code

```
def offer(row):  
    v = value(row)  
    print(f"valued at {v:,.0f} kr; the bank offers {0.8 * v:,.0f} kr")  
  
offer(house)
```

```
valued at 4,341,389 kr; the bank offers 3,473,112 kr
```

One function wraps the model, the lender's rule and the formatting. A loop over **worst.index** would write the letter for the three houses the model got most wrong.

Everything used today

Looking at a table

`read_csv`, a column's `mean` and `median`, a mask asked of every row, `groupby` for a number per town, `sort_values` and `head`, a scatter, a histogram and a line.

A split and two scores

The 2024 sales sealed until the end; RMSE in kroner, and the share of loans underwater, which is the mean of a mask.

fit and predict

`LinearRegression` on one column, two and three. Coefficients in kroner per unit, with the other columns held fixed.

Text, penalties and logs

`get_dummies` for the town and the type, a `Pipeline` with `StandardScaler`, `GridSearchCV` over alpha, the lasso's zeros, and `np.log` on a skewed target.

Loops, dictionaries, functions

The accumulator pattern over 7,621 rows, a dictionary filled by a loop, `def` with an f-string, and the vectorised line that replaces the loop.

Two rules that need no model

The average price, and a price per square metre. Every model has to beat them.

Choosing without the test rows

`KFold` for interchangeable rows, `cross_val_score`, a loop over a dictionary of candidates.

Using the model

A one-row table into `.predict()`, `.copy()` and change one column, a loop that moves the house, the largest errors by `sort_values`.

Where this goes next

before	fitting, choosing by cross-validation, and holding coefficients small	regression
now	one investigation from the raw file to a valued house, with every tool so far	a recap
next	classification, where the target is a label: will this borrower repay?	logistic regression
then	trees and ensembles, with more settings and the same grid search	the same workflow

The split, the baselines, the folds, the grid and the sealed test rows stay exactly as they were today. Only the model inside them changes.

Before the next session

- There are no exercises and no case this time. Open this deck again at home and keep changing things: value a house in your own town, make 2023 the test year, add a candidate, move the grid.
- Finish any earlier exercise set you skipped. The loops, functions and masks in the early sets are what carried every step today.
- Read the case notebooks from Part 1 to Part 6 in one sitting, as one investigation.

Stuck? Ask a friend, ask an AI for a hint rather than an answer, or email jobo@econ.au.dk.